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Underground Mining and Bulk Materials Handling (MIN E – 630)

Speech Script - Final Presentation

1 Speech Script: MIN-E-630 Final Presentation

Presenter: Jinpeng Zhu

Duration: 15 minutes

Course: Underground Mining and Bulk Materials Handling (MIN-E 630)

Instructor: Professor Wei Victor Liu

1.1 Opening [~2 min]

Good morning/afternoon, everyone.

My name is Jinpeng Zhu. I'm a graduate student in Civil and Environmental Engineering at the University of Alberta.

Today, I will present my term paper on **sensing technologies for cementitious ground support in underground mining**.

This course is called “Underground Mining and Bulk Materials Handling,” taught by Professor Wei Victor Liu.

1.2 Agenda [~1 min]

I will cover three main parts:

First, my term paper — motivation, framing, and key takeaways.

Second, beyond the paper — what I learned from writing and from this course.

Third, insights and gains from this course.

Let's begin.

1.3 About Me [~1 min]

Let me briefly introduce myself:

- I am a graduate student in Civil and Environmental Engineering.
 - My interest is in **sensing + underground mining + decision-making**.
 - Today, my focus is on **lifecycle-oriented sensing for cementitious ground support**.
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1.4 Part 1: Background & Motivation [~2 min]

My term paper title is:

“Sensing technologies for cementitious ground support systems in underground mining: advances, challenges, and future perspectives”

1.4.1 Why this topic?

When underground mines go deeper, they face:

1. **High stress** — The rock pushes hard from all sides.
2. **High temperature** — It gets hot underground.
3. **Complex water** — Groundwater causes damage.

Under these conditions, ground support becomes very important. And we can’t easily see inside the rock to check if everything is okay. That’s why we need sensors.

1.5 Part 2: Cementitious Support Types [~2 min]

Cementitious support appears in many forms:

1. **Shotcrete** — Concrete sprayed onto tunnel walls (drifts, intersections).
2. **Cast-in-place linings** — Concrete poured in place (shafts, chambers).
3. **Cemented backfill** — A mixture of tailings and cement (stopes).
4. **Grouted bolts/cables** — Bolts held in place by cement grout.

These are everywhere in underground mining. They are the backbone of safety.

1.5.1 Why is it hard to manage?

- **Performance changes with time** — It starts soft, then hardens, then degrades.
 - **Coupled hazards** — Stress, water, seismicity, and blasting all affect it together.
 - **Limited visibility** — We can’t see inside. We can’t easily verify.
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1.6 Part 3: Key Message [~2 min]

My paper does three things:

1. **Frames the problem** — Organized cementitious support monitoring into **five lifecycle phases**. Each phase asks different questions.
 2. **Links variables to decisions** — Connected what we measure (temperature, strain, cracking) to what we decide (can we re-enter? should we reinforce?).
 3. **Identifies the bottleneck** — The hardest part is not the sensors. It’s making all the data work together. Communication, data fusion, analysis, and decision-making — that’s where we need more work.
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1.7 Part 4: Five Lifecycle Phases [~4 min]

1.7.1 Phase I — Hours to Days (Placement and Early Age)

Questions: When can we re-enter safely? Did we place it well (thickness/coverage)?

We measure: Temperature → maturity → early strength

Simple example: Like baking bread — it's done when it reaches a certain temperature.

1.7.2 Phase II — Days to Weeks (Early Service)

Questions: Is it carrying load as expected? Are we seeing early cracking or interface issues?

We measure: Strain and stress

Simple example: When you lean on a table, it bends a little. Strain sensors tell us how much the support is bending.

1.7.3 Phase III — Months to Years (Long-term Degradation)

Questions: Is it degrading (creep, corrosion, cracking)? When should we maintain, reinforce, or reassess capacity?

We measure: Deformation over time, corrosion indicators

Simple example: Like a bridge that slowly sags over decades. We track this over time.

1.7.4 Phase IV — Dynamic Events (Blasting/Seismic)

Questions: Can we detect dynamic events or hazards? Can we know what has changed due to these events?

We measure: High-frequency signals (acoustic emission, microseismic)

Simple example: Like a car crash. The impact happens in milliseconds. Regular sensors are too slow.

1.7.5 Phase V — Post-event Assessment (Inspection Window)

Questions: What changed (geometry, cracking, hidden defects)? Repair, reinforce, or abandon?

We measure: Geometry, cracks, internal defects (LiDAR, photogrammetry, radar)

Simple example: Like a doctor using an X-ray to see inside.

1.8 Part 5: Technology Selection [~3 min]

Six main technology groups:

Technology	What it measures	Example
Thermal/Maturity	Temperature → strength	Thermocouples
Mechanical	Stress, strain	Strain gauges
Deformation	Movement, displacement	Extensometers
Wave-based	Cracks, damage	Acoustic emission
Durability	Corrosion, chemistry	Resistivity sensors

Technology	What it measures	Example
Geometry	Shape, size	LiDAR scanning

No single technology can do everything. We must combine them based on the phase and the question.

1.8.1 Selection Process

1. Define goals — What do we want to know?
2. Assess constraints — What's the environment like?
3. Screen technologies — Which sensors match our needs?
4. Pilot test — Try it on-site before full deployment.

1.9 Part 6: System Integration [~3 min]

1.9.1 Key Point

The sensors are not the hard part. Integration is the hard part.

The communication system is the core of the sensing system.

- Easy to deploy and install
- Strong penetration through rock
- Compatibility and anti-interference ability
- Currently, wire cable is widely used in underground mining, mainly for personnel positioning

1.9.2 Why integration is the bottleneck

1. **Constrained communications**
 - No GPS underground
 - Complex tunnel geometry
 - Metal and water block signals
2. **Heterogeneous data**
 - Point sensors give single numbers
 - Distributed sensors give lines of data
 - Visual sensors give images
 - Acoustic sensors give sound data
3. **Complex analysis**
 - Decision-making needs comprehensive understanding
 - We need to look at the big picture, not just one sensor

Simple example: Your car shows “low tire pressure” and “engine hot.” These might be related, but one sensor can't tell you that. You need a system that connects the dots.

1.10 Part 7: Challenges & Future [~3 min]

1.10.1 Challenge 1: Sensor Longevity and Robustness

Sensors must survive:

- Water and humidity

- Blasting vibration
- Installation damage
- Chemical attack from groundwater

In underground mining, the support lasts for years. But sensors often fail after months. This is still a bottleneck.

1.10.2 Challenge 2: Transition from Monitoring to Active Control

Most sites collect data but still decide by habit or experience.

We need smarter models and clear rules that connect indicators to actions.

The industry doesn't have motivation to invest in sensing systems. They only do what regulations require.

1.10.3 Challenge 3: Towards Self-Assessment and Self-Decision

In the future, systems should check themselves and make basic decisions without human help:

1. Detects abnormal strain
2. Runs a quick analysis
3. Sends an alert to the right person
4. Logs the event for future reference

This is possible with AI and automation. It's the direction of the future.

1.11 Part 8: Learning from This Course [~3 min]

Through this course, I realized that:

1. **Writing is not easy** — Inputs and ideas are the pre-requisite. Good writing needs good ideas first.
2. **Resources and support matter** — The instructor and university were critical to our learning and growth.
3. **Knowledge comes from many places** — The really important knowledge might not come from the slides. It comes from discussions and observations with others.
4. **Question industry practice** — It's not always right. It's better to understand the theory and principles behind the practice, and then make your own judgment.
5. **Challenge the status quo** — Dare to challenge things that are not reasonable.
6. **Focus on the target** — Don't just upgrade your engine. Focus on the target and notice the environment.

1.12 Closing [~30 sec]

Thank you for your attention.

To summarize:

- Sensing technologies are essential for underground mining safety.
- No single sensor works for all situations.
- The real challenge is system integration — making data useful for decisions.
- The future is automation and self-assessment.

I welcome your questions.

1.13 Timing Guide

Section	Time
Opening	2 min
Agenda	1 min
About Me	1 min
Background & Motivation	2 min
Cementitious Support Types	2 min
Key Message	2 min
Five Lifecycle Phases	4 min
Technology Selection	3 min
System Integration	3 min
Challenges & Future	3 min
Learning from Course	3 min
Closing	30 sec
Total	~15 min

1.14 Notes

Use this space for your personal notes: